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Mobile Immersive Computing: Research Challenges and the Road Ahead

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By its name, immersive computing creates a sense of immersion by blurring the boundary between the digital and the physical worlds. Typical immersive computing technologies include VR, 360-degree videos, AR, MR, and so on. The authors discuss and highlights several research challenges when consumers experience immersive media content on mobile devices.

Abstract

By its name, immersive computing creates a sense of immersion by blurring the boundary between the digital and the physical worlds. Typical immersive computing technologies include VR, 360-degree videos, AR, MR, and so on. This article discusses and highlights several research challenges when consumers experience immersive media content on mobile devices. It focuses on the recent developments for improving network efficiency and quality of user experience for mobile immersive applications. It then briefly introduces related standardization activities. Finally, it points out several opportunities and directions for further innovation, such as edge-accelerated AR applications.

Introduction

The primary functionality of immersive computing is to create a sense of immersion for its users by blurring the boundary between the physical and the simulated digital worlds. Examples of immersive computing technologies are 360° videos, Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR), and so on. Although immersive computing has recently drawn increasing attention and publicity, it is not a new concept in technology. Ivan Sutherland designed in 1968 the first Head-Mounted Display (HMD) for immersive simulation applications [1]. In 1992, the U.S. Air Force Research Laboratory created what was widely considered to be the first functioning AR application, named *Virtual Fixtures* [2].

Each immersive computing technology has its own unique features. This article focuses on three relatively mature technologies: 360° video streaming, VR and AR. 360° videos are recorded using omni-directional cameras, which allows users to freely control their viewing directions during playback. VR simulates a realistic lifelike experience and makes users feel that they are in this virtual environment. The most appealing use cases of VR are gaming and education/training. AR augments the physical world by rendering virtual content overlaid on top of real surrounding objects. It has found applications in various areas such as healthcare and communication. MR mixes real and virtual worlds and enables the interaction between physical and digital objects in real time. Figure 1 shows the so called Reality-Virtuality Continuum [3] for MR. This article also offers a brief overview of the emerging volumetric videos that enable watching the same object from different directions. When consuming volumetric videos, a viewer can not only change viewing direction along the yaw, pitch and roll axes (which 360° videos offer), but also move up and down, left and right, and back and forward via translational movements.

Recently there has been a lot of attention on immersive computing, thanks to a combination of hardware and software advancement and innovation. Regarding the hardware, numerous consumer devices have been produced by different vendors, such as Facebook Oculus Rift, HTC Vive, Sony PlayStation VR, Samsung Gear VR, Google Daydream, Microsoft HoloLens, Magic Leap One, and so on. On the software side, both Google ARCore (https://developers.google.com/ar/; accessed on 18 May 2019) and Apple ARKit (https://developer.apple.com/arkit/; accessed on 18 May 2019) were released in 2017. Google also extended VR support for its Chrome browser and integrated AR features into the same browser for downloading 3D virtual objects. Mozilla created the WebXR (https://blog.mozvr.com/tag/webxr/; accessed on 18 May 2019) APIs for accessing both VR and AR devices.

Given the low latency and high bandwidth requirements of mobile immersive applications, it is unlikely they can realize their full potential when relying on only existing computing and networking infrastructure. Thus, 5G and edge computing will play a substantial role in making these technologies a reality. In turn, mobile immersive computing may drive the wide adoption and monetization of 5G and edge computing. For example, a low motion-to-photon latency[1] smaller than 15–20 ms is necessary for mobile VR applications [4]. It is hard to achieve this low latency without the support of 5G whose target latency is 1 ms. Moreover, immersive applications push the connectivity requirement of cellular networks for offering ubiquitous access to mobile users, which is one of the key design goals of 5G (besides low latency communication).

In this article, the following section discusses several challenges of immersive media technologies when they are consumed on mobile devices, focusing on networking challenges. Then we review recent standardization activities of immersive media. We then present potential research opportunities for improving quality of user expe-

[1] Motion-to-photon latency is the time required to fully reflect a movement of a user on a display device.

Digital Object Identifier: 10.1109/MCOM.001.1800876

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rience and further innovations by exploring machine learning and mobile edge cloud. Finally we conclude the article.

Research Challenges

This section discusses several research challenges associated with 360° video streaming, AR, VR and the emerging volumetric video streaming.

360° Video Streaming

Streaming 360° videos to mobile devices is challenging, due to its high bandwidth demand. To achieve the same perceived quality, the size of 360° videos is about 5x larger than that of conventional videos. Major 360° video streaming providers, for example, YouTube and Facebook, deliver entire panoramic views, which is not network friendly as only a small portion of downloaded content is actually consumed (~15 percent for videos using Equirectangular — https://en.wikipedia.org/wiki/Equirectangularprojection; accessed on 18 May 2019 — projection).

To improve network efficiency and Quality of Experience (QoE) of 360° video streaming, viewport adaptive schemes have been proposed. They segment a video into multiple tiles and mainly send tiles overlapping with predicted viewports to users [5]. To increase the robustness of content delivery, a video player can download additional out-of-sight tiles that are adjacent to the viewport at a lower quality. To summarize, viewport-adaptive approaches deliver mainly video content inside predicted viewports (different from traditional solutions that send entire panoramic views), but they require more computation on mobile devices to decode and process multiple tiles.

A common challenge for viewport-adaptive solutions is the prediction of future viewports, which will be used to select the delivered tiles. In general, the prediction accuracy of a small time window (e.g., 100 ms) is higher than that of a large one (e.g., 500 ms). This leads to a stringent requirement on network latency. Another challenge for tile-based 360° video streaming is the rate adaptation scheme. Algorithms for non-360° video streaming consider only the time dimension and select different quality levels for different video chunks. Rate adaptations for viewport adaptive 360° video streaming also need to take the spatial dimension into consideration and choose the best quality for various tiles of the same video chunk.

Although tile-based streaming requires minimal changes on video servers, it imposes additional load on mobile devices, which need to determine the to-be-fetched tiles and then decode and stitch them together for rendering and display. In practice the decoding capability of modern mobile devices is limited by their GPU-based hardware video decoders. Thus, tile-based 360° video streaming requires powerful mobile devices to play the videos smoothly.

Edge computing brings both computation and content closer to end users for 360° video streaming. As a result, tile decoding and tile merging can potentially be performed on edge cloud. By leveraging mobile edge cloud for video transcoding, it is possible to make 360° video streaming available for all types of mobile devices, including medium and low-end smartphones. Another benefit of utilizing edge cloud is that it enables sophisticated machine learning algorithms for predicting future viewports. These algorithms in general have a higher accuracy than classic algorithms but may not be feasible to run on mobile devices (due to the high computation demand).

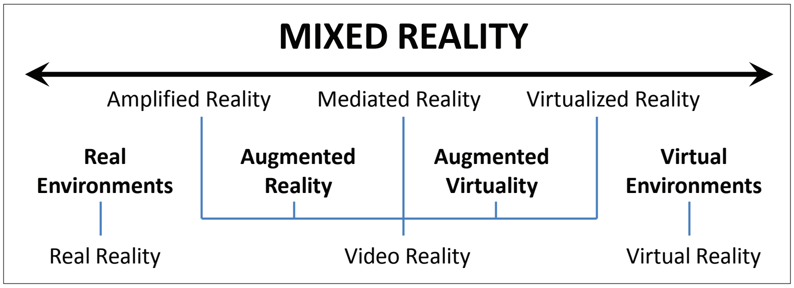

Figure 1. The Reality-virtuality continuum [3].

Figure 2 shows an edge-assisted delivery architecture for 360° video streaming with five key steps:

- Video player sends a request of video content to edge cloud along with historical viewport trajectory and the number of buffered video frames.
- Proxy at edge cloud runs machine learning algorithms to predict a future viewport based on the historical viewport movement data. If the predicted viewport has not been cached locally, proxy sends a request to download video tiles overlapping with the predicted viewport from video server.
- Video server sends the requested video tiles to proxy running on edge cloud.
- Proxy selects the best video quality based on the estimated network condition and client-side buffer occupancy. It re-generates video chunks with a different quality level if needed to enable rate adaptation. It then leverages GPU accelerated decoders to decode video tiles and combines them together to re-construct the viewport content.
- Proxy delivers a video chunk that combines viewport tiles to mobile device for rendering and display.

Augmented Reality

A crucial component in AR is to "understand" objects and there are multiple methods to achieve different levels of understanding. First, in the initial releases of ARCore and ARKit, they featured horizontal plane (e.g., table or floor) detection. Second, deep learning based methods, such as AlexNet (https://en.wikipedia.org/wiki/AlexNet; accessed on 18 May 2019) and ResNet (https://en.wikipedia.org/wiki/Residualneuralnetwork; accessed on 18 May 2019), are capable of object detection and classification. Their goal is to classify millions of images in various categories (e.g., 1000 different classes for the ImageNet — http://image-net.org; accessed on 18 May 2019 — database). However, these approaches can tell only the category of an object (e.g., a car, a flower or a chair). Third, image retrieval finds the exact match of the target object in a large-scale database of reference images, which offers rich contextual information of the recognized object for augmentation. Hence, this article focuses on the third level.

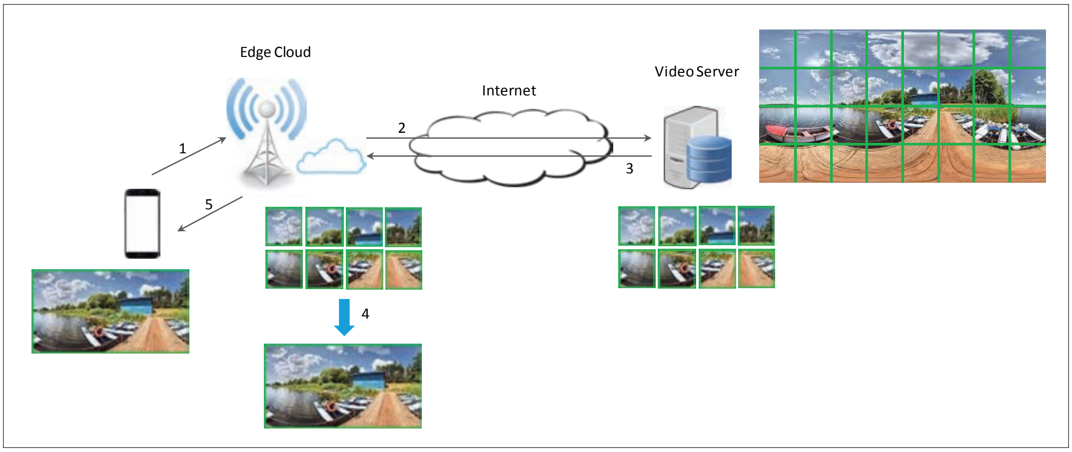

Figure 2. Edge-assisted delivery architecture for 360° video streaming with 5 steps: (1) send request to edge server; (2) forward request to video server for cache miss; (3) deliver viewport tiles to edge server; (4) combine tiles into a single video stream; and (5) deliver the combined video.

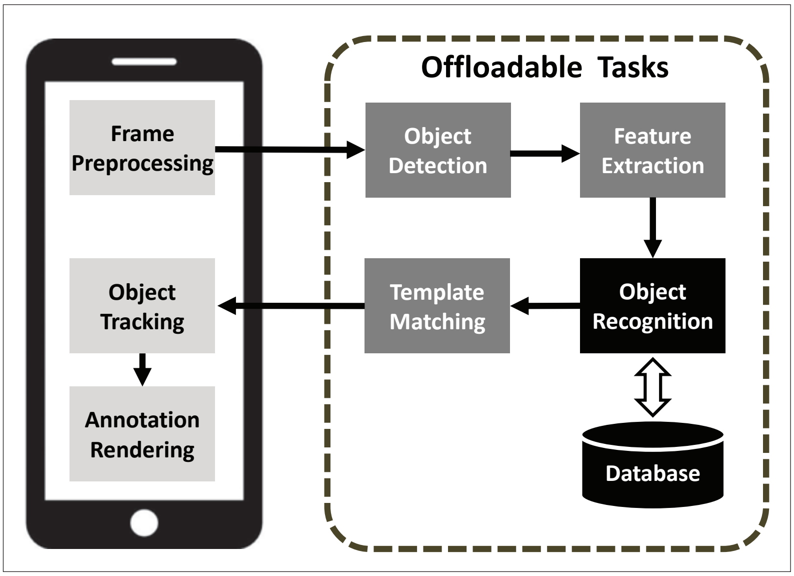

Figure 3. Pipeline of a typical mobile AR application.

Usually, image retrieval involves a complicated pipeline composed of several compute intensive algorithms, for example, object detection, feature extraction and object recognition, which makes it unrealistic to perform these tasks on mobile devices with limited computation power. Cloud offloading is a promising solution to solve this problem, where an AR system offloads the reference-image database and certain visual tasks (e.g., feature extraction and object recognition) to the cloud. Figure 3 shows the pipeline for a typical mobile AR application and illustrates the computation tasks that could be offloaded to the cloud. The offloading also changes the data model. For example, when offloading the four tasks in the dotted block, mobile devices need to send camera views to the cloud for further processing. Another option for speeding up compute intensive algorithms is to leverage recent hardware advances of mobile devices that offer various opportunities for AR without requiring cloud offloading. However, they may increase the cost of mobile devices.

Ideally, the end-to-end latency of an AR application should be lower than the camera-frame intervals, which is usually around 33 ms (i.e., 30 FPS — frames per second), in order to achieve the best quality of user experience. However, Zhang *et al*. [6] identified that even when offloading computation intensive tasks to the cloud, the server-side processing time, which is ~190 ms, still dominates the end-to-end latency. Although technologies such as optical flow tracking can be used as a remedy to track the position changes of an augmented object between consecutive camera frames, the tracking accuracy decreases when the latency to get the recognition result increases. A promising research direction for mobile AR is to reduce the end-to-end latency by leveraging the Ultra Reliable Low Latency Communications (URLLC) offered by 5G New Radio (NR).

A key networking challenge of cloud-based mobile AR applications is that they upload camera frames to the cloud for further processing which consumes uplink bandwidth. There are several existing solutions to reduce uplink data usage. For example, Glimpse [7] strategically uploads only *trigger frames* to the cloud for object recognition. It sends trigger frames under the following three conditions:

- When the local motion tracking is likely to fail (i.e., the extracted feature points deviate from their correct position)
- When the scene has changed significantly
- When there is no frame currently being processed by the cloud.

Instead of uploading camera frames, VisualPrint [8] proposes to extract visual fingerprints from these frames that are highly unique and contain the greatest global entropy and sends these fingerprints to the cloud for object recognition. However, generating fingerprints from feature points may be computation intensive and lead to more energy consumption on mobile devices and extra latency.

Besides the above image-retrieval based object recognition, another possible solution is to utilize 6DoF (six degrees of freedom) localization to recognize and track objects. MARVEL [9] is such a recent proposal that combines local inertial (https://en.wikipedia.org/wiki/Inertialnavigationsystem; accessed on 18 May 2019) and optical flow tracking with remote visual tracking in the cloud. Its goal is to enable low energy and low latency AR on mobile devices. More specifically, MARVEL mainly relies on lightweight local inertial tracking for achieving low latency processing. The local optical flow tracking is triggered when a camera image is sharp enough (i.e., the edge intensity is lower than a threshold) and it differs significantly from the previous frame which is detected by the location difference from motion sensor data. It offloads image processing to the cloud only when there is an inconsistency between inertial tracking and optical flow tracking. However, one limitation of 6DoF localization is that it requires construction of a detailed 3D model of the environment in advance and computation of feature points for an object from multiple different viewing angles to enable accurate localization.

A unique feature of AR is that by the design it ties tightly with the real world and thus its users viewing the same or overlapped scenes are in the close proximity. By leveraging the social nature of mobile AR users, it is possible to enable coordination and collaboration among them for sharing their common interests (e.g., the results of computation intensive AR tasks from the cloud, high quality AR annotations, and real-time annotation modifications). Zhang *et al.* recently proposed CARS [10], Collaborative Augmented Reality for Socialization, a collaborative AR framework that supports instantaneous socialization. CARS brings benefits at the user, application and system levels, for example, by reducing end-to-end AR latency and reusing networking and computation resources. Recently both Google's ARCore v1.2 and Apple's ARKit v2.0 enable the multiplayer feature for sharing the AR experience with someone else. It allows multiple people seeing the same object to share data among each other within a virtual environment. For the collaborative AR applications, how to address the security and privacy concerns deserves further research.

Virtual Reality

One issue of high-end VR headsets (e.g., HTC Vive and Oculus Rift) is that they are connected via cables to a powerful personal computer or game console for improving the QoE of high-quality VR applications, which limits the mobility of their players. MoVR [11] is a promising solution to solve this problem, which proposes to replace the HDMI cable with mmWave communication and addresses two key challenges of utilizing mmWave links. First, it designs a self-configurable mmWave mirror to overcome the blockage problem by reflecting wireless signals toward the receiver. Second, it relies on the tracking information in VR headsets, combined with RF-based localization, to align the receiver's beam with the transmitter's beam. However, by only replacing the cable with a mmWave link, the required data rate over this link is still extremely high, around 5 to 7 Gb/s. Moreover, mmWave is not yet available immediately for commodity mobile devices.

Note that standalone VR devices, such as Samsung Gear VR and Google Daydream, do not have the mobility constraint as they render graphics content using local compute resources. Hence, an alternative approach is to decompose the VR pipeline and move some of the computation tasks to the headsets that also have reasonable compute capability which is under-utilized in the above setup. This requires developing an in-depth understanding of the processing and resource (both compute and bandwidth) requirements at different components in the pipeline. It leads to the design of low-latency untethered VR systems. For example, Liu *et al*. [4] leverages a parallel rendering and streaming pipeline to improve end-to-end latency.

There are four key components in a typical VR pipeline: sensing, rendering, streaming and displaying. After receiving the motion sensor data from a headset on the updated viewing direction, a content server renders a new frame and streams it back to the headset for display. Without compression, the raw data rate will be ~5.6 Gb/s for headsets with a 2160×1200 resolution and 90 frames per second refreshing rate (if three bytes are used for each pixel). Given a 90Hz frame rate, the average waiting time for display is ~5.5 ms. According to Liu *et al*. [4], the time for sensing could be less than 1 ms for a VR system using a WiGig network. Depending on the workload, the rendering time varies between 5 and 11 ms. Thus, the work by Liu *et al*. [4] focuses on optimizing the streaming time, including encoding, transmission, and decoding. Specifically, to support simultaneous rendering and encoding, it takes advantage of the VR rendering nature which is separated for two eyes and sends the rendered left-eye image for compression without waiting for the rendering of the right-eye image. The same is done for the transmission and decoding on the headsets. As a result, the proposed system achieves less than 16 ms end-to-end latency for VR systems with a 2160×1200 resolution at 90Hz and 20 ms latency for 4K resolution.

	Data	Remote	Local
360° video	Panoramic video	–	Decoding
Adaptive	Viewport tiles	Segmentation	Tile merging
AR	–	–	Entire pipeline
Offloading	Camera view	Object recognition	Object tracking
VR	Raw data	Rendering	Display
Untethered	Encoded data	Encoding	Decoding

Table 1. Summary of the consumed data and computation requirements of immersive technologies.

Table 1 summarizes the above immersive technologies, in terms of their consumed data and computation requirements, and compares different solutions for each technology. For traditional 360° video streaming, the local computation on a client is mainly the decoding of panoramic frames. Viewport-adaptive approaches need to decode and merge viewport tiles that are generated by servers via segmentation. For traditional AR applications, the entire pipeline is executed by mobile devices. During offloading, an AR system sends camera views to remote cloud servers that perform tasks such as object recognition. When untethering VR headsets from backend computers, instead of sending raw data to display on the client, the server delivers encoded data to reduce bandwidth consumption.

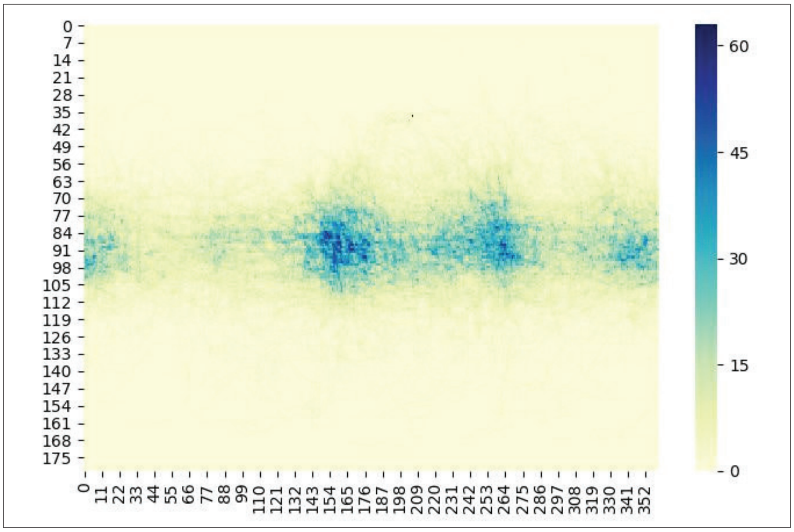

Figure 4. Heatmap of a 360° video. The x axis is the longitude and the y axis is the latitude.

Emerging Volumetric Video Streaming

Immersive media content is evolving with emerging technologies such as volumetric videos, which lead to various new research directions. Different from 360° videos which are captured *inside-out*, volumetric videos are created *outside-in*, which enables watching the same object from different angles. When watching 360° videos, a user can freely change viewing direction along the yaw, pitch and roll axes through rotational movements, which is not enough to support a fully immersive experience. Volumetric videos also enable a viewer to move up and down, left and right, and backward and forward via translational movements. Hence, they are actually the logical next step after stereoscopic 360° videos.

A key concept of volumetric video streaming is voxel, which adds depth information to pixel in a 3D space [12]. A frame of a volumetric video may consist of millions of voxels. Thus, volumetric videos are much more demanding in terms of network bandwidth (with data rate potentially higher than 100 Mb/s) than that of 360° videos. This makes volumetric video streaming more challenging and requires the support from high-bandwidth network design, such as enhanced Mobile BroadBand (eMBB) in 5G NR.

One potential solution for improving bandwidth efficiency is to extend tile-based streaming for 360° videos to cube-based streaming for volumetric videos, by transmitting only visible cubes to viewers. However, viewport prediction is more difficult for volumetric video streaming, due to the increased dimensions of movement that viewers are allowed to, compared to 360° videos (from 3 to 6). Another promising direction is to leverage edge cloud for effective video transcoding, which will be presented later.

With the increasing popularity of immersive media content, point cloud (i.e., a set of points in a 3D space, defined by their locations and attributes, for example, color and opacity) has been a potential enabler for reconstructing an object or a scene. It is usually captured using multiple cameras with depth sensors and may have up to billions of points for a realistic reconstruction. Hence, point cloud compression has been an active research area for both AR and volumetric videos. It can support a wide range of use cases, including real-time 3D immersive telepresense, 3D free viewport sports broadcasting, autonomous navigation based on large-scale 3D dynamic maps, and so on. Both objective and subjective quality assessment of point cloud compression is still an open research problem (e.g., how to define rate distortion metrics and how to guarantee the stable/comparable viewing between test subjects).

Standards Activities

This section presents recent standardization activities on immersive media technologies from MPEG (the Moving Picture Experts Group), 3GPP (the 3rd Generation Partnership Project), VRIF (Virtual Reality Industry Forum), and DVB (Digital Video Broadcasting).

MPEG has defined ISO/IEC 23090 (which is also called MPEG-I — https://mpeg.chiariglione.org/standards/mpeg-i; accessed on 18 May 2019), *Coded Representation of Immersive Media*. It currently has eight parts: Technical Report on Immersive Media describes the scope of this standard; Omnidirectional Media Format is an application format for omnidirectional video media (a.k.a. OMAF, Omnidirectional Media Application Format) and it defines the encapsulation, signaling and streaming of omnidirectional videos for Dynamic Adaptive Streaming over HTTP (DASH); and the remaining six parts are Versatile Video Coding, Immersive Audio Coding, Point Cloud Compression, Immersive Media Metrics, Immersive Media Metadata, and Network Based Media Processing (NBMP). NBMP is a framework that describes in-network media processing operations defined by service providers and end users for enabling interoperability. It aims to standardize service function chaining of network-based media processing functions. It offers control functions for composing and configuring media processing and transport methods for communications between media processing entities.

3GPP has published TR 26.918 on *Virtual Reality Media Services over 3GPP* (http://www.3gpp.org/dynareport/26918.htm; accessed on 18 May 2019), which currently focuses on 360° video and the associated audio. TR 26.918 collects comprehensive information on various use cases for VR, including event broadcast/multicast, live services consumed on HMD, social TV and VR, learning applications, VR calls, and so on. It also presents the test methodology, experimental setup and results for both video and audio quality evaluation. For example, one listening experiment explains how to derive the detection threshold for motion-to-sound latency. VRIF has four working groups on the *Requirements* and *Guidelines* on accessibility, interoperability, quality and deployment of the VR ecosystem, *Liaison* to other industry groups and *Communications* to its members and public at large. It covers almost all aspects of the VR ecosystem, such as content capture, production, distribution, presentation and security. DVB has established the Commercial Module CM-VR working group to draft commercial requirements for VR systems.

Research Opportunities

This section discusses some research opportunities for innovations of immersive computing, which are enabled by machine learning and mobile edge computing. It also explores the opportunities related to quality of experience.

Machine Learning

Machine learning will play a key role in enabling mobile immersive computing. It is possible to predict future viewports using machine learning models based on the historical viewport trajectory. The reason is that the viewport for 360° videos is determined directly by, for example, the head orientation of viewers when using head mounted displays. In order to confirm the predictability of future viewports, Qian *et al*. [5] performed a large-scale user study with more than 130 participants watching 10 popular 4K 360° videos. As shown in Fig. 4, the heatmap of content access patterns generated from the collected traces demonstrates that there are focused areas for these videos. Qian *et al*. [5] also showed that even with simple machine learning algorithms (i.e., linear regression) it is possible to achieve a reasonable prediction accuracy.

Viewport prediction accuracy can be improved by combining other data sources with head movement traces. For example, a data-fusion model can jointly consider cross-user viewing statistics of the same video or single-user viewing behaviors over multiple videos, video content analysis (e.g., saliency map, heatmap and motion tracking) and content analysis of spatial audios, and contextual information from users (e.g., their engagement levels). This model can use ensemble learning (https://en.wikipedia.org/wiki/Ensemblelearning; accessed on 18 May 2019) to combine multiple learning algorithms with different data sources and improve prediction accuracy. Another promising research direction is to leverage deep learning models, such as Long Short-Term Memory (LSTM), to provide more accurate viewport prediction. As mentioned above, mobile edge cloud has been an ideal platform to run these compute-intensive deep learning algorithms [13].

In order to guarantee an acceptable quality of user experience, the resolution of a 360° video should be at least 4K. 8K or even 16K resolution is recommended for achieving a truly interactive and immersive experience. Such a high resolution leads to huge bandwidth consumption when streaming 360° videos over mobile networks. Recent work by Yeo *et al.* [14] demonstrated that deep neural networks can fundamentally change how videos are delivered over the Internet. The proposed content-aware super-resolution recovers a high resolution video frame from a low resolution one by utilizing a deep convolution network. Applying similar ideas to improve the network efficiency of 360° video streaming, especially for videos with a high frame rate (e.g., 90 or even 120 FPS) is still an open research problem.

Edge Computing

Besides assisting 360° video streaming, mobile edge cloud may facilitate other immersive applications, for example, the delivery of volumetric videos to resource-constrained mobile devices. Directly delivering uncompressed volumetric videos to mobile devices is bandwidth consuming. The bitrate of a volumetric video with around 50K points per frame on average could be as high as 100 Mb/s. Although point cloud compression can reduce network data usage, decoding the compressed videos on mobile devices is challenging. Even for a high-end smartphone, the achievable FPS is around only 3–4, much lower than the desired 24–30 FPS. Mobile edge cloud with servers equipped with GPUs offers an ideal venue to alleviate the high decoding demand on end-user devices.

In this client/server model, edge servers are responsible for transcoding video content in the predicted viewport from point cloud format to pixel-based regular videos, which reduces the computation overhead of decoding, rendering and display on mobile devices. In order to improve the QoE when viewport prediction is not accurate, edge servers can also deliver a low quality point cloud stream to the client along with the transcoded pixel-based video stream which represents only the viewport content. Mobile devices will display the transcoded pixel-based video and the low quality point cloud stream.

When offloading part of the image retrieval pipeline to the edge cloud, hardware acceleration using GPUs can be leveraged to optimize the end-to-end latency of mobile AR applications. GPU acceleration for several image-retrieval tasks, for example, feature extraction and feature-point encoding, has been investigated before in a standalone manner. However, the question of whether the accelerated pipeline could satisfy the above stringent latency requirement of mobile AR systems has not yet been meaningfully addressed. Zhang *et al*. recently proposed Jaguar [6], an edge cloud based AR system that features accurate, low-latency and large-scale object recognition. Jaguar further pushes the limit of end-to-end AR latency and reduces it to ~33 ms (thanks to GPU acceleration). However, for cameras with a high FPS (e.g., 60 or 90) and time-critical applications such as remote surgery, it is expected that the end-to-end latency should be further decreased.

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User Experience

Although there is a plethora of work on improving network efficiency of 360° video streaming, little has been done to understand the user experience when watching 360° VR videos. To fill this gap, Van den Broeck *et al.* [15] investigated how users interact with 360° VR content on a smartphone, a tablet and an HMD. Based on a measurement study of 18 participants watching six 360° videos, they found that the HMD provides the most immersive experience at the cost of motion sickness and a higher cognitive burden. The above study demonstrated that the ergonomics of HMD is an essential pillar for offering a truly immersive VR experience. For example, an HMD should not be too heavy. Otherwise, it will affect head balance and body posture and create a mismatch between visual and proprioceptive cues. The temperature inside an HMD is another key factor which if not handled properly will break the VR experience. To avoid light leakage, headsets are usually sealed tightly around the user's face, Measuring the quality of user experience for immersive computing, especially for 360° video streaming, is still an under-explored research topic, although there is a rich literature on the subjective methodologies and objective tools for the quality assessment of regular video streaming.

which makes the temperature control an important problem for HMD design.

Measuring the quality of user experience for immersive computing, especially for 360° video streaming, is still an under-explored research topic, although there is a rich literature on the subjective methodologies and objective tools for the quality assessment of regular video streaming. For example, the number of visual quality switches between adjacent chunks is an important QoE metric for traditional video streaming (frequent jumps of visual quality may degrade QoE). For tile-based 360° video streaming, it is still not clear what is the best metric to quantify the impact of quality switches between tiles in the same chunk, as not all tiles are equal (e.g., the quality switch of tiles around the viewport boundary may have less impact on QoE). Although visual quality is important, it alone is not enough to measure the QoE for mobile immersive applications. Other factors include auditory, interactive and ergonomic experience, which all affect the physical comfort of users (e.g., dizziness, eye strain, nausea, and so on.).

Conclusion

This article reviewed several immersive computing technologies, including virtual reality, 360° video streaming, augmented reality, and so on, by discussing the networking challenges when enabling these technologies on mobile devices. It then introduced recent standardization activities of immersive media technologies. After that, it presented several future research directions on leveraging machine learning and edge cloud for improving the performance and QoE of mobile immersive applications. Given its interdisciplinary feature, more initiatives may emerge from not only the networking research community, but also other related disciplines, to tackle various challenging issues introduced by mobile immersive computing.

Acknowledgment

The author thanks the anonymous reviewers for their insightful comments, and Feng Qian, Wenxiao Zhang and Vijay Gopalakrishnan for discussions and suggestions.

References

[1] I. E. Sutherland, "A Head-Mounted Three Dimensional Display," *Proc. AFIPS*, 1968, pp. 757–64.
[2] L. B. Rosenberg, "Virtual Fixtures: Perceptual Tools for Tele robotic Manipulation," *Proc. IEEE Annual Int'l. Symposium on Virtual Reality*, 1993, pp. 76–82.
[3] P. Milgram and F. Kishino, "A Taxonomy of Mixed Reality Visual Displays," *IEICE Trans. Information and Systems*, vol. 77, no. 12, 1994, pp. 1321–29.
[4] L. Liu *et al.*, "Cutting the Cord: Designing a High-quality Untethered VR System with Low Latency Remote Rendering," *Proc. ACM MobiSys*, 2018, pp. 68–80.
[5] F. Qian *et al.*, "Flare: Practical Viewport-Adaptive 360-Degree Video Streaming for Mobile Devices," *Proc. ACM MobiCom*, 2018, pp. 99–114.
[6] W. Zhang, B. Han, and P. Hui, "Jaguar: Low Latency Mobile Augmented Reality with Flexible Tracking," *Proc. ACM Multimedia*, 2018, pp. 355–63.
[7] T. Y.-H. Chen *et al.*, "Glimpse: Continuous, Real-Time Object Recognition on Mobile Devices," *Proc. ACM SenSys*, 2015, pp. 155–68.
[8] P. Jain, J. Manweiler, and R. Roy Choudhury, "Low Bandwidth Offload for Mobile AR," *Proc. ACM CoNEXT*, 2016, pp. 237–51.
[9] K. Chen *et al.*, "MARVEL: Enabling Mobile Augmented Reality with Low Energy and Low Latency," *Proc. ACM SenSys*, 2018, pp. 292–304.
[10] W. Zhang *et al.*, "CARS: Collaborative Augmented Reality for Socialization," *Proc. ACM HotMobile*, 2018, pp. 25–30.
[11] O. Abari *et al.*, "Enabling High-Quality Untethered Virtual Reality," *Proc. USENIX NSDI*, 2017, pp. 531–44.
[12] J. Park, P. A. Chou, and J.-N. Hwang, "Rate-Utility Optimized Streaming of Volumetric Media for Augmented Reality," *IEEE J. Emerging and Selected Topics in Circuits and Systems*, vol. 9, no. 1, 2019, pp. 149–62.
[13] Y. He *et al.*, "Software-Defined Networks with Mobile Edge Computing and Caching for Smart Cities: A Big Data Deep Reinforcement Learning Approach," *IEEE Commun. Mag.*, vol. 55, no. 12, 2017, pp. 31–37.
[14] H. Yeo, S. Do, and D. Han, "How will Deep Learning Change Internet Video Delivery?" *Proc. ACM HotNets*, 2017, pp. 57–64.
[15] M. V. den Broeck, F. Kawsar, and J. Schöning, "It's All Around You: Exploring 360° Video Viewing Experiences on Mobile Devices," *Proc. ACM Multimedia*, 2017.

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